

Chapter 2

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Exercise 1.

Show that if a *finite* group G acts on a Hausdorff space Y such that none of its elements has a fixed point, the action is even and thus it yields a Galois cover $Y \rightarrow G \backslash Y$ for connected Y .

Proof.

Pick some $y \in Y$. As the space is Hausdorff, for every $gy \neq y$ there exists disjoint open sets U_1, U_g of y and gy respectively. Then as G is finite, we obtain a finite collection of open sets $\{U_g\}$, which are pairwise disjoint open neighbourhoods of the gy 's. Then as the collection of open sets is necessarily finite we can take the intersection $\bigcap g^{-1}U_g$ to obtain an open neighbourhood of y whose G -orbit is pairwise disjoint.

It remains to show that distinct elements of G produce distinct open sets: It is sufficient to show that $gy \neq g'y$ for all $g \neq g'$; as the Hausdorff condition would imply that $U_g \neq U_{g'}$. But then if such a g, g' exists, the element $gg'^{-1} \neq 1$ would fix the point y . ■

Exercise 2.

Show that a connected cover $Y \rightarrow X$ is Galois with automorphism group G if and only if the fibre product $Y \times_X Y$ is isomorphic to the trivial cover $Y \times G \rightarrow Y$ (where G carries the discrete topology).

Proof.

Recall that $Y \times_X Y$ can be characterized as $\{(y, y') \in Y \times Y : \pi(y) = \pi(y')\}$. So each point of $Y \times_X Y$ above y is exactly the set $\pi^{-1}(y) \times \pi^{-1}(y)$. For the following proof, we will understand $Y \times_X Y$ to be a Y -cover via $\pi_X(y, y') = y$.

$(\Rightarrow)$: Suppose that $Y \times_X Y$ is isomorphic to $Y \times G$ as Y -covers via some ϕ . Clearly we have that G acts simply transitively on $Y \times G$ by multiplication on the right coordinate, so we can define an induced action of G on $Y \times_X Y$ via pre-composing with ϕ . As ϕ is a homeomorphism the induced action is also necessarily simply transitive.

$(\Leftarrow)$: Suppose that Y is Galois. Then G acts transitively on the fibers, such that $\pi_X^{-1}(y)$ is equal to the set $\{(y, gy)\}_{g \in G}$. Hence we get a well-defined bijection $[\phi : Y \times_X Y \rightarrow Y \times G]$.

We check that ϕ is continuous: Let $U \times V$ be a basic open of $Y \times G$. Then $\phi^{-1}(U \times V) = \{(y, gy) : y \in U, g \in V\} = \bigcup_{g \in G} U \times_X gU$, which is seen to be open.

We check that ϕ is open: Let $U \times_X U'$ be a basic open set of $Y \times_X Y$. Then $\phi(U \times_X U')$ is seen to be a basic open set in $Y \times G$ as well. Since G carries the discrete topology we only need verify the image be open on the Y -coordinate. But then that is exactly U . ■

Exercise 3.

Let X be a connected and locally simply connected topological space and $x \in X$ a fixed base point. Consider a connected cover $p : Y \rightarrow X$ and a base point $y \in p^{-1}(x)$.

(a) Show that there is a natural injective homomorphism $\pi_1(Y, y) \rightarrow \pi_1(X, x)$.

Proof.

Let $f : [0, 1] \rightarrow Y$ be a loop at y . Then by direct composition with the covering map we obtain a loop $p \circ f$ at $p(y) = x$. That this defines a group homomorphism is a routine verification.

Now suppose that $f \in \pi_1(Y, y)$ such that $p \circ f$ is the constant path $\{x\}$. Then the image of f can only land inside $\pi^{-1}(x)$, which is a discrete space. But then the continuous image of a connected set is still connected, so f is necessarily constant; i.e. the homomorphism is injective. ■

(b) Viewing $\pi_1(X, x)^{\text{op}}$ as a subgroup of $\text{Aut}(\tilde{X}_x | X)$ via the isomorphism $\text{Aut}(\tilde{X}_x | X) \cong \pi_1(X, x)^{\text{op}}$, establish an isomorphism $\pi_1(Y, y)^{\text{op}} \setminus \tilde{X}_x \cong Y$.

Proof.

Recall (Proposition 2.4.6) that the isomorphism $\pi_1(X, x) \cong \text{Aut}(\tilde{X}_x | X)^{\text{op}}$ is given by $\alpha \mapsto \phi_\alpha$, where $\phi_\alpha(f) := f \circ \alpha$. This also gives an isomorphism $\pi_1(Y, y) \cong \text{Aut}(\tilde{X}_x | Y)^{\text{op}}$. But then as the embedding $\pi_1(Y, y) \hookrightarrow \pi_1(X, x)$ is defined via composition with the covering map, it necessarily agrees with the isomorphism as the group homomorphism $\text{Aut}(\tilde{X}_x | Y) \rightarrow \text{Aut}(\tilde{X}_x | X)$ is defined via composition with the covering map. So $\pi_1(Y, y)^{\text{op}} \setminus \tilde{X}_x \cong \text{Aut}(\tilde{X}_x | Y) \setminus \tilde{X}_x \cong Y$. ■

Exercise 4.

Let G be a connected and locally simply connected topological group with unit element e . Let $\pi : \tilde{G}_e \rightarrow G$ be a universal cover, and $\tilde{e}$ the universal element in the fiber $\pi^{-1}(e)$.

(a) Equip $\tilde{G}_e$ with a natural structure of a topological group with unit element $\tilde{e}$ for which π becomes a homomorphism of topological groups.

Proof.

Recall (Construction 2.4.1) that the universal cover $\tilde{G}_e$ is constructed as the set of paths starting at e up to (fixed end point) homotopy. Hence, we define a composition law on $\tilde{G}_e$ as follows: given $f, g : [0, 1] \rightarrow G$, we let $fg(t) := f(t)g(t)$. That this is indeed a well-defined path follows from the continuity of the group operation and connectedness of G . That this defines a group law with identity element $\tilde{e}$, that π is a group homomorphism and that the diagram below commutes is a routine verification. It remains to show continuity of the group operations.

$$\begin{array}{ccc} \tilde{G}_e \times \tilde{G}_e & \xrightarrow{\tilde{m}} & \tilde{G}_e \\ \pi \times \pi \downarrow & & \downarrow \pi \\ G \times G & \xrightarrow{m} & G \end{array} \qquad \begin{array}{ccc} \tilde{U} \times \tilde{V} & \xrightarrow{\quad} & \tilde{W} \\ \downarrow & & \downarrow \\ U \times V & \xrightarrow{\quad} & W \end{array}$$

Suppose that $\tilde{f}, \tilde{g} \in \tilde{G}_e$ such that $\tilde{f}\tilde{g} = \tilde{h}$. Let $\tilde{W}$ denote some open simply connected neighbourhood of $\tilde{h}$. It is then sufficient to show that $(\tilde{f}, \tilde{g}) \in \tilde{G}_e \times \tilde{G}_e$ admits a basic open neighbourhood contained in $\tilde{m}^{-1}(\tilde{W})$. To that end, by the continuity of multiplication in G , there exists $(f, g) \in U \times V \subseteq m^{-1}(W)$. By shrinking the neighbourhoods U, V if necessary we can WLOG assume they are simply connected and a local trivialization of $\pi \times \pi$. Hence by taking an appropriate lift there exists $(\tilde{f}, \tilde{g}) \in \tilde{U} \times \tilde{V}$.

It is then sufficient to show that $\tilde{m}(\tilde{U} \times \tilde{V}) \subseteq \tilde{W}$. To that end, note that points in $\tilde{U} \times \tilde{V}$ are of the form $(\tilde{\alpha} \circ \tilde{f}, \tilde{\beta} \circ \tilde{g})$ where $\tilde{\alpha}, \tilde{\beta}$ are paths contained in $\tilde{U}, \tilde{V}$ with startpoints at $\tilde{f}(1)$ and $\tilde{g}(1)$ respectively. Then $\tilde{m}(\tilde{\alpha} \circ \tilde{f}, \tilde{\beta} \circ \tilde{g}) = \tilde{\alpha}\tilde{\beta} \circ \tilde{f}\tilde{g}$. But then $\alpha\beta$ is a path contained in $U \times V$, such that its image is contained in W . By passing through the lift π , we see that the path $\tilde{\alpha}\tilde{\beta}$ must be contained in $\tilde{W}$, and we are done.

The proof for showing that inversion is a continuous operation is near-identical, and we omit it here for the sake of brevity. ■

(b) Show that a group structure with the above properties is unique.

Proof.

Suppose that m, m' are two group structures on $\tilde{G}$ that satisfy the given requirements. Then the following must be a commuting diagram of topological groups:

$$\begin{array}{ccc} \tilde{G} \times \tilde{G} & \xrightarrow{\quad m' \quad} & \tilde{G} \\ & \searrow \quad m \quad & \downarrow \pi \\ & & G \end{array}$$

but then we are immediately done by Proposition 2.2.2., as $\pi(m'(\tilde{e}, \tilde{e})) = \pi(m(\tilde{e}, \tilde{e})) = e$. ■

(c) Construct an exact sequence of topological groups

$$1 \rightarrow \pi_1(G, e)^{\text{op}} \rightarrow \tilde{G}_e \rightarrow G \rightarrow 1,$$

the topology on $\pi_1(G, e)$ being discrete.

Proof.

From (a) we have that π and the embedding of $\pi_1(G, e)^{\text{op}}$ into $\tilde{G}_e$ are well-defined group homomorphisms. As a covering map it is necessarily surjective; and it is immediate that its kernel is $\pi_1(G, e)^{\text{op}}$. We verify that the SES agrees with the given topologies: It is sufficient to show that the topology endowed on $\pi_1(G, e)^{\text{op}}$ as a subspace of $\tilde{G}_e$ is discrete. But as then there is a natural identification of the points in $\pi_1(G, e)^{\text{op}}$ with the elements of $\text{Aut}(\tilde{G}_e | G)$ (Proposition 2.4.6); and the action of $\text{Aut}(\tilde{G}_e | G)$ on $\tilde{G}_e$ is even (Proposition 2.2.3), we see that the induced subspace topology on $\pi_1(G, e)$ is exactly the discrete topology. ■

Exercise 5.

Let $p : Y \rightarrow X$ be a cover of a connected and locally simply connected topological space X . For a point $x \in X$, we have defined two canonical left actions on the fibre $p^{-1}(x)$: one is the action by $\text{Aut}(Y|X)$ and the other is that by $\pi_1(X, x)^{\text{op}}$. Verify that these two actions commute, i.e. that $\alpha(\phi(y)) = \phi(\alpha y)$ for $y \in p^{-1}(x)$, $\phi \in \text{Aut}(Y|X)$ and $\alpha \in \pi_1(X, x)$. [*Hint*: Use Theorems 2.3.5 and 2.3.7.]

Proof.

By Theorem 2.3.7. there is a natural identification of the action of $\text{Aut}(\tilde{X}|X)$ on $p^{-1}(x)$ with the action of $\pi_1(X, x)^{\text{op}}$. But then by Theorem 2.2.10. $\text{Aut}(Y|X)$ is naturally identified with the coset space of $\text{Aut}(\tilde{X}|X) / \text{Aut}(\tilde{X}|Y)$. Hence the two actions can be identified with the same group action, such that they commute. ■

Exercise 6.

Let $\tilde{X}_\Delta \rightarrow X$ be the cover of the connected and locally simply connected space X introduced in Remark 2.4.14 (2).

(a) Verify that the $\pi_1(X, x)$ -space corresponding to $\tilde{X}_\Delta$ via Theorem 2.3.4 is $\pi_1(X, x)$ acting on itself via inner automorphisms.

Proof.

Recall that $\tilde{X}_\Delta$ is defined as the pullback with respect to the diagonal map $X \rightarrow X \times X$ and the cover $\tilde{X} \rightarrow X \times X$ (where $\tilde{X}$ denotes the fundamental groupoid of X). In more concrete terms, the fibre of $\tilde{X}_\Delta$ above $x \in X$ are homotopy classes of loops based at x , with the coarsest topology such that the projection is continuous.

As in the construction of the universal cover (2.4.1), we can give an open neighbourhood basis for $\tilde{x}$ as follows: For $x \in X$, let U be some simply connected neighbourhood of x . Then the basic open set $\tilde{U}$ is the set of classes of loops formed by composing $\alpha \in \pi_1(X, x) = \pi^{-1}(x)$ on both sides with some path ϕ in X from x to $y \in U$ to obtain $\phi \circ \alpha \circ \phi^{-1} \in \pi_1(X, y) = \pi^{-1}(y)$.

Then as the monodromy action is obtained via lifting a path along successive simply connected neighbourhoods, we obtain that the monodromy action is exactly $\phi \bullet \alpha = \phi \circ \alpha \circ \phi^{-1}$; i.e. inner automorphism. ■

(b) Let $f : [0, 1] \rightarrow X$ be a path with endpoints $x = f(0)$ and $y = f(1)$. The pullback $f^* \tilde{X}_\Delta$ is a trivial cover of $[0, 1]$ by simply connectedness of $[0, 1]$, whence an isomorphism of fibres $\tilde{X}_{\Delta, x} \xrightarrow{\sim} \tilde{X}_{\Delta, y}$. Verify that under the identifications $\tilde{X}_{\Delta, x} \cong \pi_1(X, x)$ and $\tilde{X}_{\Delta, y} \cong \pi_1(X, y)$ this isomorphism corresponds to the isomorphism of fundamental groups $\pi_1(X, x) \xrightarrow{\sim} \pi_1(X, y)$ induced by $g \mapsto f \bullet g \bullet f^{-1}$.

Proof.

By chasing the pullback diagram, one finds that $f^* \tilde{X}_\Delta$ is the subspace of $\tilde{X}_\Delta \times [0, 1]$ consisting of points of the form $(\pi_1(X, f(t)), t)$; with maps being the projection maps onto each coordinate.

Given any arbitrary cover Y of $[0, 1]$ and some point $p \in \pi^{-1}(0)$, consider taking the lift of the identity path $\varphi : [0, 1] \rightarrow [0, 1]$ along p (ie the monodromy action). Then the connected component of p is exactly the image of the lifted path $\tilde{\varphi}_p$, and this allows us to identify the isomorphism $Y_0 \cong Y_1$.

Hence, in the specific case of $f^* \tilde{X}_\Delta$, we find that the isomorphism between $\pi_1(X, x)$ and $\pi_1(X, y)$ is explicitly given by $\alpha \mapsto \tilde{\varphi}_\alpha \bullet \alpha$; which we know to be equal to $\tilde{\varphi}_\alpha \circ \alpha \circ \tilde{\varphi}_\alpha^{-1}$ from the result of part (a). It remains to show that $\tilde{\varphi}_\alpha$ is exactly f . To that end, consider the following commutative diagram:

$$\begin{array}{ccccc}
 & & f^* \tilde{X}_\Delta & \longrightarrow & \tilde{X}_\Delta \\
 & \nearrow \tilde{\varphi}_\alpha & \downarrow & & \downarrow \\
 [0, 1] & \xrightarrow{\varphi_\alpha} & [0, 1] & \xrightarrow{f} & X
 \end{array}$$

where the composite map $[0, 1] \rightarrow \tilde{X}_\Delta$ is exactly the lift of the path f ; which from (a) we know to be the conjugated path $t \mapsto f(tx) \circ \alpha \circ f(tx)^{-1}$. But then this lifts along the connected component of $(\alpha, 0)$ to give us that $\tilde{\varphi}_\alpha \bullet (\alpha, 0) = f \circ (\alpha, 0) \circ f^{-1}$. ■

Exercise 7.

(Deligne) Let X be a connected and locally simply connected topological space. A *groupoid cover* over X is a cover $\pi : \Pi \rightarrow X \times X$ together with the following additional data. Denote by s and t the maps $\Pi \rightarrow X$ obtained by composing π with the two projections $X \times X \rightarrow X$, and consider the fibre product $\Pi \times_X \Pi$ with respect to the map s on the left and t on the right. The groupoid structure is given by three morphisms of spaces over $X \times X$: a multiplication map $m : \Pi \times_X \Pi \rightarrow \Pi$, a unit map $e : X \rightarrow \Pi$ (where X is considered as a space over $X \times X$ by means of the diagonal map), and an inverse map $\iota : \Pi \rightarrow \Pi$ which satisfies $t \circ \iota = s$, $s \circ \iota = t$. These are subject to the following conditions:

$$m(m \times \text{id}) = m(\text{id} \times m), \quad m(e \times \text{id}) = m(\text{id} \times e) = \text{id},$$

$$m(\text{id} \times \iota) = e \circ t, \quad m(\iota \times \text{id}) = e \circ s.$$

A morphism of groupoid covers is a morphism of covers of $X \times X$ compatible with the above additional data.

(a) Verify that the space $\tilde{X} \rightarrow X \times X$ of Construction 2.4.13 carries the structure of a groupoid cover over X .

Proof.

The multiplication map m is composition of paths $(f, g) \mapsto f \circ g$, which is well-defined as the fibered product is taken with respect to the projection maps s and t . The unit map e is the diagonal embedding that sends a point to the constant loop at that point. The inverse map is the map that sends a path to the reverse path. That these definitions satisfy the conditions for a groupoid cover is a routine verification and is omitted here. ■

(b) Fix a base point (x, x) of $X \times X$ coming from a point $x \in X$. Show that the cover $\tilde{X} \rightarrow X \times X$ corresponds via Theorem 2.3.4 to the underlying set of $\pi_1(X, x)$ together with the left action of $\pi_1(X, x) \times \pi_1(X, x)$ given by $(\gamma_1, \gamma_2)\gamma = \gamma_1\gamma\gamma_2^{-1}$.

Proof.

First notice that $\pi_1(X \times X, (x, x)) = \pi_1(X, x) \times \pi_1(X, x)$ as there is a natural bijection $\text{Hom}([0, 1], X \times X) \cong \text{Hom}([0, 1], X) \times \text{Hom}([0, 1], X)$. Next, notice that $\pi^{-1}(x, x)$ is by definition the collection of homotopy classes of paths from x to x , such that it is directly identified with the set $\pi_1(X, x)$. That the induced monodromy action is exactly as stated is another routine verification. ■

(c) Show that for every groupoid cover $\Pi \rightarrow X \times X$ there is a unique morphism of groupoid covers $\tilde{X} \rightarrow \Pi$.

Proof.

Skipped.

Exercise 8.

Let X be a connected and locally path-connected, but not necessarily locally simply connected topological space. Construct a profinite group G such that the category of finite covers of X becomes equivalent to the category of finite sets equipped with a continuous left G -action.

Proof.

Notice that if Y, Z are two finite covers of X , then the (fibered) product $Y \times_X Z$ is a finite cover of X via Y and Z of degree $\deg(Y) \deg(Z)$. Hence the category of finite covers of X form a filtered inverse system of topological spaces. If in particular one assumes that every finite cover is dominated by a finite Galois cover $(*)$, then by the correspondence of Theorem 2.2.10. we will have that the inverse system of finite covers of X is isomorphic to the inverse system of corresponding automorphism groups. In particular, the inverse limit of the system will be the group G as desired, and the proof of the question statement is analogous to the proof found in Corollary 2.3.9.. Hence we will dedicate the remainder of this proof to proving $(*)$:

Fix some base point $x \in X$, and WLOG suppose that Y is a connected cover of degree n such that $p^{-1}(x) = y_1, \dots, y_n$. Now notice that the n -fold product Y_X^n is a degree n^n cover of X , such that it admits an embedding $S_n \hookrightarrow \text{Aut}(Y_X^n | X)$ via permutation of the coordinates (it is a well-defined deck transformation as coordinate permutations are isomorphisms of $Y^n \supseteq Y_X^n$).

Let $\hat{Y}$ denote the connected component of the n -fold product Y_X^n that contains the point $(y_1, \dots, y_n) \in Y \times_X \dots \times_X Y$. Note that as the coordinates are distinct, this space admits the same action of S_n . As $\hat{Y}$ is connected by assumption, the monodromy action of $\pi_1(X, x)$ on $\hat{Y}$ induces a morphism $\phi : \pi_1(X, x) \rightarrow S_n \rightarrow \text{Aut}(\hat{Y}|X)$ such that the image $H := \phi(\pi_1(X, x))$ is a transitive subgroup of S_n . But then the degree of $\hat{Y}$ is exactly equal to the size of the $\pi_1(X, x)$ orbit on the fiber (as connected, locally path connected spaces are necessarily path connected), so ϕ is in fact surjective on $\text{Aut}(\hat{Y}|X)$. We conclude that $\hat{Y}$ is Galois. To finish, notice that $\hat{Y}$ is a finite cover of Y via projection onto the first coordinate. ■

In fact we prove something stronger: That the Galois cover $\hat{Y}$ above is terminal among all Galois covers of Y , such that it is indeed deserving of the name “Galois closure”:

Suppose that Z is some Galois cover of X such that its projection factors through a cover morphism $f : Z \rightarrow Y$. Fix some z_0 such that $f(z_0) = y_1$, and choose $g_i \in \text{Aut}(Z|X)$ such that $f(g_i z_0) = y_i$ (take $g_1 = e$). Then we can define a morphism $\Psi : Z \rightarrow Y_X^n$ to be given by $z \mapsto (f(g_1 z), \dots, f(g_n z))$ such that $\Psi(z_0) = (y_1, \dots, y_n) \in \hat{Y}$. By the connectedness of Z we see that $\Psi(Z) \subseteq \hat{Y}$, and by Lemma 2.2.11. we see that Ψ is in fact a covering map.

Exercise 9.

Let S be a set having at least two elements, and let X be a topological space. Define a presheaf $\mathcal{F}_S$ of sets on X by setting $\mathcal{F}_S(U) = S$ for all nonempty open sets $U \subset X$ and $\rho_{UV} = \text{id}_S$ for all open inclusions $V \subset U$. Give a necessary and sufficient condition on X for $\mathcal{F}_S$ to be a sheaf.

Proof.

The author has made the strange choice in defining sheaves that two sections need only be glued together when they agree on a *nonempty* intersection (Definition 2.5.4.). I will be ignoring this convention.

We show that the necessary and sufficient condition is for every pair of (non-empty) open sets to have nonempty intersection.

Necessary: Suppose to the contrary that $U \cap V = \emptyset$ for nonempty open sets U, V . Then as $|S| \geq 2$, pick $s, s' \in S$ distinct. Then the sections $s \in \mathcal{F}_S(U)$ and $s' \in \mathcal{F}_S(V)$ cannot be glued to a section on $U \cup V$, so $\mathcal{F}_S$ cannot be a sheaf.

Sufficient: Suppose we have a collection of sections (s_i, U_i) that agree on intersections. We show that this implies that every section must in fact be the same element of S : Choose two sections s_i, s_j . Then $U_i \cap U_j \neq \emptyset$, so $s_i = s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} = s_j$. But then $U_i \cap U_j \subset U_i$ and $U_i \cap U_j \subset U_j$ so we get that $s_i = s_j$. ■

Exercise 10.

Let X be a topological space. Show that the category of sheaves on X is equivalent to the category of those spaces $p : Y \rightarrow X$ over X where the projection p is a local homeomorphism (i.e. each point in Y has an open neighbourhood such that $p|_U$ is a homeomorphism onto its image). [*Hint*: Begin by showing that for a sheaf $\mathcal{F}$ the projection $p_{\mathcal{F}} : X_{\mathcal{F}} \rightarrow X$ is a local homeomorphism.]

Proof.

We prove this via the construction of a functor and its inverse. We first show that $\mathcal{F} \mapsto X_{\mathcal{F}}$ is a well-defined functor:

Let $\mathcal{F}$ be some sheaf over X . Let $s = [(U, t)]_x \in \mathcal{F}_x$ denote some arbitrary point of $X_{\mathcal{F}}$. Then we have an associated induced map $i_s : U \rightarrow X_{\mathcal{F}}$, which is continuous by definition, which we see to be a homeomorphism between U and $i_s(U)$. Indeed, $i_s \circ p_{\mathcal{F}} = p_{\mathcal{F}} \circ i_s = \text{id}$.

Now suppose that $\Phi : \mathcal{F} \rightarrow \mathcal{G}$ is some morphism of sheaves. Then define a map $\phi : X_{\mathcal{F}} \rightarrow X_{\mathcal{G}}$ to be given by $[(U, t)]_x \mapsto [(U, \Phi(t))]_x$. That this is indeed well-defined follows from the compatibility definition of a morphism. That this map commutes with the respective projections onto X follows from construction. Finally it is continuous as $\phi^{-1}(i_{\Phi(s)}(U)) = i_s(U)$.

We next show that $Y \mapsto \mathcal{F}_Y$ is a well-defined functor:

That $\mathcal{F}_Y$ is a well-defined sheaf follows from Proposition 2.5.8.. Now suppose that $\Psi : Y \rightarrow Z$ is a morphism of étale spaces over X . Then for U open in X , define $\psi : \mathcal{F}_Y(U) \rightarrow \mathcal{F}_Z(U)$ to be given by $s \mapsto \Psi \circ s$. As this map is given by left composition of a function, that is automatically guaranteed to satisfy the compatibility condition on restrictions, so it induces a well-defined morphism of sheaves.

We now show that $\mathcal{F} \mapsto \mathcal{F}_{X_{\mathcal{F}}} \cong \text{id}$:

To $s \in \mathcal{F}(U)$, we associate the section $i_s : U \rightarrow X_{\mathcal{F}}$. This induces a well-defined morphism of sheaves, as restriction on sections in $\mathcal{F}(U)$ correspond to restricting the domain of i_s . As $\mathcal{F}$ is assumed to be a sheaf, differing sections must have at least one differing germ; such that this association is injective. Now let $i : U \rightarrow X_{\mathcal{F}}$ be a section with $V \subseteq i(U)$ mapped homeomorphically down to $p(V)$. Then $i^{-1}(V) = p^{-1}(V)$, such that $i|_V$ is a local homeomorphism. So the association is surjective and an isomorphism. That this defines a natural transformation is a routine verification.

We finally show that $Y \mapsto X_{\mathcal{F}_Y} \cong \text{id}$:

For $y \in Y$, let V be a neighbourhood of y that is mapped homeomorphically down to U . Then to y we associate the germ $[(U, s)]_{p(y)}$, where $s : U \rightarrow V$ is the corresponding section. This association is seen to be injective, as differing y will always either have distinct stalks or germs. It is also seen to be surjective, as $[(U, t)]_x = [(U, t)]_{p(y)}$ always for some $p(y)$, where we shrink U if necessary to obtain the section $t : U \rightarrow V \ni p(y)$ as needed. Finally we see that this association defines a natural transformation in Y , as morphisms $Y \rightarrow Z$ induce morphisms $\mathcal{F}_Y \rightarrow \mathcal{F}_Z$ via post-composition of sections; and the induced morphism $X_{\mathcal{F}_Y} \rightarrow X_{\mathcal{F}_Z}$ can be seen to commute with the resultant diagram. ■

Exercise 11.

Let $\mathcal{F}$ be a presheaf of sets. Show that there is a natural morphism of presheaves $\rho : \mathcal{F} \rightarrow \mathcal{F}_{X_{\mathcal{F}}}$, and moreover every morphism $\mathcal{F} \rightarrow \mathcal{G}$ with $\mathcal{G}$ a sheaf factors as a composite $\mathcal{F} \xrightarrow{\rho} \mathcal{F}_{X_{\mathcal{F}}} \rightarrow \mathcal{G}$.
*[Remark: For this reason $\mathcal{F}_{X_{\mathcal{F}}}$ is called the *sheaf associated with $\mathcal{F}$* .]*

Proof.

We have that any morphism of presheaves $\mathcal{F} \rightarrow \mathcal{G}$ induces a morphism $X_{\mathcal{F}} \rightarrow X_{\mathcal{G}}$ of spaces over X in a functorial manner (see p.50). In particular, by using the equivalence of categories proved in Exercise 10, this in turn induces a morphism $\mathcal{F}_{X_{\mathcal{F}}} \rightarrow \mathcal{F}_{X_{\mathcal{G}}}$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & \mathcal{G} \\ \downarrow & & \downarrow \\ \mathcal{F}_{X_{\mathcal{F}}} & \longrightarrow & \mathcal{F}_{X_{\mathcal{G}}} \cong \mathcal{G} \end{array}$$

where $\mathcal{F}_{X_{\mathcal{G}}} \cong \mathcal{G}$ by the equivalence. ■